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Harnessing generative AI in chemical engineering education: Implementation and evaluation of the large language model ChatGPT v3.5

Matthew Keith [a,*], Eleanor Keiller [b], Christopher Windows-Yule [a], Iain Kings [a], Phillip Robbins [a]

[a] School of Chemical Engineering, University of Birmingham, Edgbaston Park Road, Birmingham B15 2TT, United Kingdom
[b] Wolfson Institute of Population Health, Queen Mary University of London, Mile End Road, London E1 4NS, United Kingdom

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ABSTRACT

With the recent and rapid growth of the adoption of generative artificial intelligence (GAI), including the use of large language models (LLMs) there has been growing concern amongst higher education institutions regarding assessment, potential plagiarism, and ultimately a negative impact on student learning outcomes. However, GAI is likely to be a useful tool in future professional environments, including in many chemical engineering-related roles. It is, therefore, essential that students are equipped with the knowledge and skills to use GAI responsibly, ethically, and safely. This research adopts the IDEE (Identify desired outcomes, Determine level of automation, Ensure ethics, Evaluate effectiveness) framework to develop a chemical engineering lab session which is augmented by the use of LLMs. As part of the pre-lab work, Year 1 students were tasked with using ChatGPT v3.5 to derive a model which predicted the drainage profile of water from a tank. They then tested the validity of this model experimentally in a lab session and analysed the data obtained as part of the post-lab work. Pre- and post-lab surveys were conducted which revealed that students had limited prior experience with GAI but there was a general belief that it could be useful for future work. The post-lab survey showed that the vast majority of people believed that this exercise had helped them learn how to use LLMs, how to use it ethically, how to critique the output, and what some of its limitations were. Reflexive thematic analysis was applied to the qualitative data obtained in the same surveys. This revealed eight distinct themes, one of which showed that there was a strong awareness of the need for criticising the LLM output, of the potential pitfalls associated with its use, and concerns over the quality of the output. As such, this work provides not just a case study for the integration of LLMs, and GAI more broadly, into chemical engineering curricula, but also valuable insight into student perceptions regarding the use of this nascent technology more generally.

1. Introduction

1.1. Background to generative artificial intelligence (GAI)

November 2022 saw the release of ChatGPT v3.5 by OpenAI and, with it, access to generative artificial intelligence (GAI) exploded into the public domain. Other large language models (LLMs) such as Gemini (Google), Perplexity AI, and WordAi (Nomi) quickly followed which allowed the fast and easy generation of high volumes of human-like text (Kanbach et al., 2024). These LLMs, and many other tools such as image generators (e.g. DALL-E), audio generators (e.g. Murf AI), and code generators (e.g. Codiga), are all based on the principles of machine learning (ML) where a computer can adapt its output without being specifically programmed to perform a pre-defined outcome (Gilliam, 2018). A wide variety of algorithms are used to achieve this, including supervised, unsupervised, and reinforcement learning, along with pattern recognition (Wakefield, 2024). Although the machine is trained on specific data sets and can “learn”, the predictive text output is purely based on statistics. Hence, it lacks subject-specific understanding, the ability to reason, and critical thinking skills (Arnold and Jantke, 2024) which are of fundamental importance to higher education (HE) degree-level programmes.

1.2. GAI in higher education

There is, understandably, a significant level of concern regarding the use of GAI within HE settings. Firstly, with regards to LLMs in particular, there is the risk of plagiarism within written coursework (Chan, 2023)

* Corresponding author.
E-mail address: m.j.keith@bham.ac.uk (M. Keith).

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which, in turn, could undermine academic integrity and the reputation of universities (Cotton et al., 2024). Subsequently, this may devalue a degree from the perspective of future graduates' employers (Holmes et al., 2022). Over-reliance on GAI may also lead to a reduction in students' critical thinking skills and ability to consolidate information from a range of sources (Warschauer et al., 2023). Coursework is regarded as a more authentic form of assessment (Villarroel et al., 2020), however, the widespread use of GAI may drive universities towards more closed-book exams and thus reduce the opportunity for students to develop these transferrable skills. There is, therefore, a risk to education quality and a potentially negative impact on student learning outcomes (Chan and Lee, 2023). Due to the issue of open-book assessments being vulnerable to cheating, and closed-book assessments being less authentic, there is, a need to develop alternative assessment formats. These could incorporate GAI and still enable students to develop those transferrable skills, which this work aims to do.

Notwithstanding the potential risks to education posed by GAI, there are several arguments that this technology can benefit education which have been summarised by (Moorhouse et al., 2023). Careful assessment design may promote cognitive ability (Bahroun et al., 2023), although other research has found that most improvements due to GAI occurs at the lower levels of Bloom's taxonomy (Essien et al., 2024). From an educator perspective, LLMs in particular are likely to be beneficial in terms of providing assessment questions, enhancing student support, and automating tailored feedback (Celik et al., 2022). The advent of GAI is clearly disruptive, but other technological innovations such as calculators, search engines, online encyclopaedias, and smartphones were too, and yet the education sector largely adapted to incorporate their use (Moodie, 2016). It is true that GAI is different to these inventions: it can neither be effectively banned nor detected (Kassorla, 2023), but, like online encyclopaedias and smartphones, it will not simply disappear. There is, therefore, a growing need to teach the ethical and responsible use of GAI to ensure that future graduates are well prepared for a workplace in which GAI is ubiquitous. This has been acknowledged by international organisations such as UNESCO (UNESCO, 2022), and by the Quality Assurance Authority (QAA) within the UK (QAA, 2023). Furthermore, a recent EduCAUSE poll found that 83 % of higher education stakeholders believed that "generative AI will profoundly change higher education in the next three to five years" (McCormack, 2023). It is, therefore, evident that universities must rise to the challenges and opportunities presented by GAI.

Fortunately, a wide range of guidance has been published to ensure the safe, ethical, and responsible incorporation of GAI into higher education curricula. These have been published by numerous educational bodies such as the QAA (QAA, n.d.), various universities (Moorhouse et al., 2023), the Alan Turing Institute (The Alan Turing Institute, 2024), and the UK government (Department for Education, 2023). One example from the Higher Education Futures Institute (HEFi) emphasises the need for students to acknowledge AI generated content, but also provides policy on assessment design (Higher Education Futures Institute, 2023). Further, numerous reflections on achieving GAI literacy, practice examples and student outputs have recently been published (Abegglen et al., 2024) with some making the case that the use of GAI should be introduced to curricula early on in degree programmes (McGowan, 2024). Additionally, some frameworks have been published to facilitate the introduction of GAI into education. For example, the IDEE framework (on which the teaching and learning activity described herein is based) consists of four key steps (Su and Yang, 2023), which are outlined below.

1. Identify the Desired Outcomes: Identify the objectives of the application to ensure that the use of technology aligns with the learning outcomes.
2. Determine the Appropriate Level of Automation: Depending on the objectives, decide whether the teaching and learning activity should be fully automated or supplementary to traditional activities.
3. Ensure Ethical Considerations: Carefully consider ethical implications, including potential biases, their impact on educators and students.
4. Evaluate the Effectiveness: Following the teaching and learning activity, evaluate how effective GAI was in achieving the learning outcomes.

Thus far, this guidance is general to higher education, and so the following section summarises the available literature on GAI specifically within chemical engineering degree programmes.

1.3. GAI in chemical engineering education

Since the late 1990s, chemical engineering education has evolved to incorporate a growing variety of computer software. From the first integration of Aspen Plus (Wankat, 2002), additional programmes such as Matlab (Sun and Zhao, 2023), COMSOL (Herrmann et al., 2022), and Simulink (Moodley, 2020), have been shown to enhance core chemical engineering knowledge, develop critical thinking skills, and prepare students for work in an increasingly digitalised world, often referred to as Industry 4.0 (dos Santos et al., 2018). In such a world where AI may be ubiquitous, designing alternative assessments where students can evaluate an AI output is desirable (Ravi, 2023). The development and incorporation of LLMs into curricula may then be viewed as another technique for the future engineer's toolbox. Indeed, it has been noted that, by utilising LLMs, chemical engineering students may be able to go beyond the original questions, and thus gain a more holistic understanding of the given problem. However, the same work does acknowledge that outputs "must be carefully evaluated for accuracy" (Tsai et al., 2023). A key challenge of teaching the use of GAI is demonstrating this need to students, and thus using AI to enhance their critical thinking skills.

If the value of a degree lies in preparing students for a professional workplace, then the need to verify the output is essential, especially with regards to ethical practice. For example, ChatGPT was used to write a valve isolation procedure (McDermid, 2023), but for this procedure to be used ethically, it must be checked by a competent engineer to ensure compliance with existing site operations and regulations. In the same trial, ChatGPT also demonstrated that it is capable of fabricating information for a RIDDOR (Reporting of Injuries, Diseases and Dangerous Occurrences Regulations) report (McDermid, 2023), meaning that the causes, remedial actions, and subsequent learning opportunities were false. This fabrication, known as AI hallucination, has previously been highlighted as a major ethical risk, likely with legal implications (Ji et al., 2023).

By carefully evaluating the solutions provided by GAI, students can identify errors, propose improvements, and hence develop a deeper understanding of specific concepts, whilst becoming more reflective and critical of their own work (Tsai et al., 2023). In this paper, this school of thought has been used to integrate GAI, specifically the widely available LLM ChatGPT v3.5, into a chemical engineering lab session with the aim of demonstrating its potential uses and pitfalls, and hence illustrate the need to be able to critique the output of LLMs. The remainder of this paper briefly summarises the role of lab sessions in chemical engineering education, before describing the teaching and learning activities delivered, and analysing the feedback collected from students before and after these activities.

1.4. Lab sessions in chemical engineering education

Across many subjects, lab sessions play a crucial role in higher education. They enhance learning outcomes, offer opportunities for authentic assessment, promote collaboration, and encourage the development of transferable skills (Dickens and Arlett, 2008). It is an example of experiential learning, and so students are able to deepen their understanding of fundamental concepts, gain practical experience

(Tormey et al., 2021), and hence improve their awareness of safety protocols and procedures; all factors which are essential to chemical engineering graduates. Conversely, previous pedagogical research has found that there may be a lack of coherent learning objectives in lab sessions (Feisel and Rosa, 2013). Limited resources or lab space frequently drives a need for small groups and hence the lab session is delivered throughout a semester. In turn, this may lead to a lack of synchronisation between taught lecture content and the experiment itself (Edward, 2002). Often in higher education, lab sessions are delivered by post-graduate research students (post-graduate teaching assistants, PGTAs) who may lack a subject specific knowledge and an awareness of the wider undergraduate course, and so direct links to other topics may be lost (Okolie et al., 2020). Within the lab session itself, poor learning is likely due to "insufficient activation of the prehension dimension" i.e. the fundamental concepts of the session are not adequately understood (Abdulwahed and Nagy, 2013) which may be addressed by the introduction of pre- and post-lab work. An awareness of these potential risks means that mitigation strategies can be developed, as discussed in Section 2.

There are numerous studies investigating the use of GAI (Alasadi and Baiz, 2023; Michel-Villarreal et al., 2023), teaching and leaning methods (Koh and Doroudi, 2023; Tsai et al., 2023), and staff and student perspectives (Kelly et al., 2023; Malik et al., 2023; Smolansky et al., 2023). Methods for integrating GAI into chemical engineering degrees are only just being explored (Tsai et al., 2023) and so this field is still in its infancy. Hence, there is a need to develop effective educational strategies which enhance learning outcomes (e.g. by being able to effectively evaluate information) whilst teaching the ethical, responsible, safe, and legal use of GAI. Linked to this ethical use, is developing an understanding in students of the enormous computational power (and associated energy use) of such tools (Berthelot et al., 2024), especially when a large proportion of students claim to study chemical engineering due to their concerns regarding climate change (Falomo and Keith, 2024).

Due to their open-ended nature, it is essential that students develop and use critical thinking skills during the planning, execution, and data analysis stages of their lab sessions. Lab sessions, therefore, offer an opportunity into which GAI education may be effectively integrated. The research presented herein describes the development, implementation, and evaluation of an undergraduate chemical engineering lab session with a specific focus on using LLMs.

2. Teaching and learning activities

2.1. IDEE framework

Related to the ideas of McGowan (2024), it was deemed most appropriate to introduce students to GAI early in the chemical engineering degree programme. Hence, the lab session targeted was for first year students and delivered in the first semester, but after they had experienced two traditional lab sessions. This introduced university lab environments before including the use of an additional technology. To develop the teaching and learning activity, the IDEE framework (described in Section 1.2) was applied, with a summary provided in Table 1.

Table 1
Application of the IDEE framework to this chemical engineering lab session.

Step	Framework action	Application
1	Identify desired learning outcomes	Three learning outcomes (LOs) relevant to the use of GAI were identified: LO1: Students should be able to develop an understanding of how to use GAI ethically, including the use of prompt engineering to generate accurate models. LO2: Students should be able to criticise the output of a large language model and thus identify and discuss potential pitfalls and limitations. LO3: Students should be able to identify improvements to the output and hence use the model to predict the experimental data.
2	Determine level of automation	As this lab session also aimed to develop practical skills in data analysis, an understanding of fluid dynamics, and experience in mathematical modelling, it was agreed by discussion amongst the authors that the level of automation should be supplementary to the main activity. Thus, a pre-lab worksheet was developed which incorporated GAI, but also specifically asked non-AI-augmented questions.
3	Ensure ethical considerations	The key ethical considerations identified were: 1. Inclusivity: it was mandated that the free online version of ChatGPT v3.5 should be used. All students would have access to this using University computers if they do not have a personal computer. 2. Data protection: Students were given the option of using staff-generated accounts with random username / password pairs if they would prefer not to use their own email addresses. 3. Biases: LLMs can exhibit biases based on training data and so students were encouraged to critically evaluate the outputs. 4. Academic integrity: Using ChatGPT as a learning tool was emphasised in a pre-lab lecture and the associated assignment was marked on the basis of students' understanding and skills, not just on the use of the technology. 5. Informed consent: A lecture (which was recorded) was used to deliver information regarding how GAI works, it's ethical use, and potential risks. 6. Transparency: To ensure that ChatGPT v3.5 was used, screenshots showing the input prompts and output were required as part of the assignment.
4	Evaluate effectiveness	The effectiveness of whether the stated learning outcomes were met was evaluated through the completion of pre- and post-lab surveys, hosted on the University's virtual learning environment (VLE).

2.2. Lab session description

The aim of this lab session was for students to develop an understanding of fluid dynamics as fluid drained from a tank under gravity. In advance of the session, all materials (the lecture slides and recording, lab brief, pre-lab worksheet, and post-lab worksheet) were made available to students via the University of Birmingham's virtual learning environment (VLE). In the lab, two sets of the same equipment were available to use. This consisted of two plastic tanks mounted to a wooden board through a load cell. The load cell was connected to a PhidgetBridge (which converts the signal from the load cell into a readable digital output). This was connected to a laptop via a USB cable and the change in mass of each tank as it drained was logged in real-time. Tank 1 has an internal diameter of 65 mm and Tank 2 an internal diameter of 100 mm. Both were filled with water to a height of 135 mm. The base of both tanks contained a hole into which a 3D printed poly(lactic acid) (PLA) plug could be inserted. This plug contained a single hole through which the water drained. For Tank 1, a circular hole, 5 mm in diameter was used. As part of the pre-lab work, students were tasked with drawing a plug in OpenSCAD to fit to the base of Tank 2 with a hole which enabled this larger tank to drain in the same time as Tank 1. This exposed students to both computer-aided design (CAD) and open-source software as part of the same piece of work. A diagram of this experimental set-up and associated plugs are provided in Fig. 1.

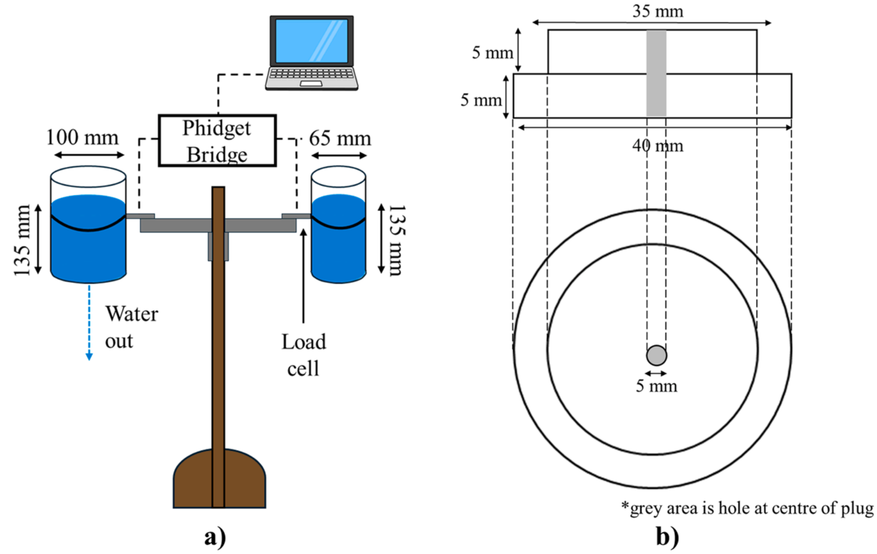

Fig. 1. Diagram of a) experimental set up of tank draining lab, and b) plug inserted into base of the tank, which students were asked to modify.

In the lab itself, students were shown how to use the software. The output of the load cell (in volts, V) was highlighted and the principle of drawing a calibration curve was explained. A jug containing ~1.5 L of water was weighed, water was poured into Tank 1 to a fill height of 135 mm, and the jug weighed again. The difference between the two measurements was taken to give the initial mass of water in Tank 1. Data logging at a rate of 4 Hz was started and a rubber bung was removed from the hole within the plug until the tank was fully drained. This procedure was repeated with Tank 2 which contained the 3D printed plug drawn by one of the students in each lab group. A minimum of four total experimental runs (two per tank) were completed by each group. At the end of the session, data were exported from the Phidget software to a.csv file. This was saved to a USB memory stick for later analysis by the students in the post-lab work.

2.3. Fluid dynamics theory

The following section aims to outline the relevant theory which was demonstrated as part of this lab session. Similar material was delivered to students in the weeks prior to the lab session as part of the module "Introduction to Transport Phenomena". As a fluid drains from a tank under gravity, a mass balance approach can be applied to predict the change in mass (or height) of the fluid over time. A diagram of the system analysed here is provided in Fig. 2, with Eqs. (1) to (12) illustrating the analytical approach taken to develop a model.

Starting from the general expression of a mass balance (Eq. (1)), the input, generation, and consumption terms are all equal to zero, hence Eq. (1) reduces to Eq. (2).

$$Input + Generation = Output + Accumulation + Consumption \tag{1}$$

$$Output = \quad - Accumulation \tag{2}$$

The output term is the volumetric flow rate of the fluid leaving through the orifice (diameter, d) multiplied by the fluid density. As an unsteady state problem, the accumulation term can be taken as a differential change in mass (dM) per differential time (dt). Substituting these terms into Eq. (2) gives Eq. (3).

$$Q\rho = \quad -\frac{dM}{dt} = \quad -\rho A\frac{dh}{dt} \tag{3}$$

Assuming that the cross-section of both the tank and the orifice remains constant, then substituting the relevant equations for the area of a circle into Eq. (3) gives Eq. (4). In turn, this can be simplified to Eq. (5) which gives an expression for the rate at which the fluid height changes as a function of the velocity of the fluid leaving through the orifice.

$$\rho u \quad \frac{\pi d^2}{4} dt = \rho\frac{\pi D^2}{4} dh \tag{4}$$

$$\frac{dh}{dt} = u \quad \frac{d^2}{D^2} \tag{5}$$

The velocity of the fluid leaving the orifice, u_2, can be determined through the application of Bernoulli's principle between points 1 and 2 in Fig. 2. This is derived in Eqs. (6) to (9) below, and assumes that the pressure at points 1 and 2 is the same ($P_1 = P_2$), that the velocity of the surface at point 1 is very small compared to the velocity at point 2 (i.e. $u_1 \approx 0$), and that the height of the fluid at point 2 is 0 m (i.e. the tank is fully emptied). Note that to account for irregular flow patterns and friction between the fluid and the orifice, it is necessary to include a discharge coefficient, C_V, which describes the deviation of real flow from ideality.

$$P_1 + \quad \frac{1}{2}\rho u_1^2 \quad + \rho g h_1 = \quad P_2 + \quad \frac{1}{2}\rho u_2^2 \quad + \rho g h_2 \tag{6}$$

$$\rho g h_1 = \quad \frac{1}{2}\rho u_2^2 \tag{7}$$

$$u_2 = \sqrt{2gh_1} \tag{8}$$

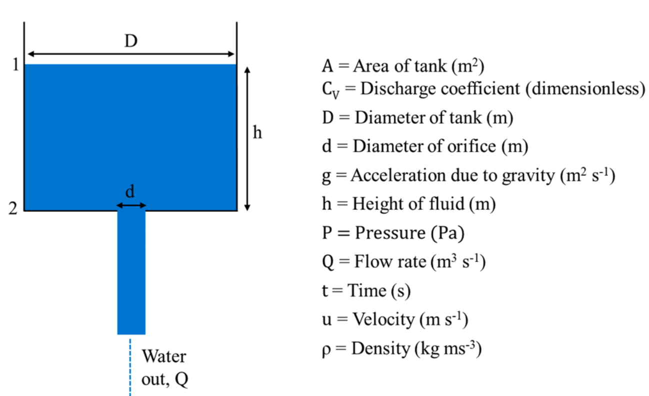

Fig. 2. System diagram and associated nomenclature.

$$u_2 = \quad C_V\sqrt{2gh_1} \tag{9}$$

This is, of course, an oversimplification; u_2 will change as the height of the fluid changes, however, it is suitable as an approximate model for a first-year undergraduate lab session. Substituting Eq. (9) into Eq. (5) yields Eq. (10). Separating the variables and integrating between the limits $h = h_1$ at $t = 0s$ and $h = h$ at $t = t$ s, then gives an expression to predict the fluid height in the tank at any time, t (Eq. (11)). By incorporating the density of the fluid and area of the tank, Eq. (12) can be developed to describe how the mass of fluid changes with time and hence used to predict the experimental data recorded as part of the lab session.

$$\frac{dh}{dt} = C_V\sqrt{2gh_1} \quad \frac{d^2}{D^2} \tag{10}$$

$$\sqrt{h} = \sqrt{h_1} - \quad \sqrt{2g}\,\frac{C_V}{2}\frac{d^2}{D^2}t \tag{11}$$

$$\sqrt{m} = \frac{D}{2}\sqrt{\pi\rho}\left(\sqrt{h_1} - \quad \sqrt{2g}\,\frac{C_V}{2}\frac{d^2}{D^2}t\right) \tag{12}$$

2.4. Pre-lab work

A two-hour lecture was delivered with the aim of giving students an introduction to GAI and LLMs. The lecture took an active-learning approach and encouraged engagement through direct questions, opinion polls, and collaborative working to develop example input prompts. Initially, the statistically methods underpinning LLMs were introduced, before giving some real-world examples of where LLMs regurgitate false or illogical information. Following this, ethics concerning the use of GAI were introduced which focussed on the application of FAST (Fairness, Accountability, Sustainability, Transparency) principles. Related to this is the need for proper referencing which was highlighted and exampled following guidance published by Springer Nature.(Springer Nature, 2023). The lecture then explicitly linked the use of GAI to chemical engineering applications and explored the need for human intervention to maintain quality and ensure safety. Finally, the lecture demonstrated the use of ChatGPT v3.5 using student-generated input prompts to different engineering problems and concluded with a description of the lab activity and associated coursework. In this way, it was hoped that students gained greater understanding of how GAI, and LLMs in particular, work and insight into how to use these tools, before attempting their own use as part of the coursework. However, the author's experience of this lecture was that the discussion was often led by a small number of individuals who were willing to contribute. It may therefore be appropriate to lead an additional GAI tutorial session in which students can be engaged and assisted on a small-group basis.

Students were given two weeks to complete a pre-lab worksheet. This asked them to make a prediction of what a graph of water mass vs. time would look like as a tank drained under gravity, derive a model which predicted the draining time using ChatGPT v3.5, and criticise the output. Students were guided in the use of prompts for other engineering problems to develop ideas on effective prompt engineering such that ChatGPT v3.5 could give a more detailed and precise output. Students were also tasked with deriving it themselves using a mass balance to evaluate whether the mathematical logic developed by ChatGPT v3.5 was correct and where to identify where the output could be improved. The primary examples of this (as captured in the Supplementary Materials) were through providing more detailed assumptions, exploring their impact and / or validity, deriving Torricelli's law from first principles, and clearly numbering and referring to equations. This also represented a key learning opportunity as it demonstrated the difficulty of distinguishing between facts and an AI hallucination. As such, the highest marks were awarded to students who derived the model separately from ChatGPT v3.5 and then compared their derivation to the ChatGPT output. This was how learners discriminated the LLM output in the clearest manner. In the second part of the worksheet, students were asked to use their model to calculate what the drainage time should be. In most cases, the ChatGPT v3.5 output matched the mass balance derived output. Where it differed, the choice of which model to use was not prescribed. The final section asked students to use their model to calculate the diameter of the hole needed to drain Tank 2 in the same time as Tank 1, and hence provide a 3D drawing of a plug with that hole. Students were not required to use ChatGPT v3.5 for this section, however, if prompted correctly, this LLM could rearrange the model for hole diameter. This pre-lab sheet was assessed and worth 20 % of a 10-credit module. This weighting was applied based on University of Birmingham guidance for the quantity of work expected for a 10 credit module. Within the pre-lab sheet, the model developed and use of LLMs accounted for 40 % of the available marks. This was the largest single portion of the pre-lab work, yet the weighting was deliberately chosen such that if a student was not able to demonstrate sufficient understanding of how to use an LLM and develop a model, they were unlikely to achieve higher than 2(ii) in this work.

To address equality, diversity and inclusion (ED&I) concerns, the use of ChatGPT v3.5 was mandated as this is freely available through internet browsers, either using University computers or personal laptops. To mitigate against potential data protection concerns, random email / password pairs were created and distributed to students who preferred not to use either a personal or University email address to create an OpenAI account.

The role and depth to which ChatGPT v3.5 was used varied across the student cohort. At the simplest level (which was not scored highly) students simply copied and pasted the lab brief question into ChatGPT v3.5, and similarly directly used the output. As shown in the Supplementary Materials, Table S1, this resulted in an incomplete and incorrect model; the shape and size of the outlet orifice was not considered and it was not able to determine suitable boundary conditions to complete the integration. Advances in this approach modified the input prompt to include specific phrases aimed at tailoring the output. For example, including "As a chemical engineer…", resulted in the output referring specifically to fluid dynamics and taking a more structured approach (see Table S2 in Supplementary Materials). Specific requests, such as a list of assumptions used, a clear definition of nomenclature and the inclusion of their units, and the use of numbered equations also improved the initial output (Supplementary Materials Table S3). Some students went further and continued to prompt Chat GPT v3.5 querying it's responses and seeking greater clarification, for example, the origin of Torricelli's Law. An example of this is provided in the Supplementary Materials Sable S3.

2.5. Lab session delivery

After submitting the pre-lab sheet, students worked in small groups of two to four people with no more than two groups completing the lab at the same time. Teaching was delivered by academic staff or a PGTA who had been trained in the weeks prior to the lab session. To allow sufficient time for explanation, demonstration of the equipment, and repeat runs, one hour time slots were scheduled for each pair of lab groups. The lab was conducted on Fridays in Weeks 6–8 of the first semester.

2.6. Post-lab work

The post-lab work focussed on analysing the experimental data and comparing it to the model derived with the aid of ChatGPT v3.5. It also asked them to discuss the likely impact of any assumptions made, how the model could be improved, and finally how these improvements could be verified. Like the pre-lab sheet, this element of the course was also worth 20 % of a 10-credit module. Students were all assigned the same hand-in date in Week 11 (the final week of the semester) and so they had between 3 and 5 weeks to complete the coursework. Here the grading policy was designed such that if students demonstrated that they met the learning outcomes of the lab (e.g. by plotting graphs comparing a model and experimental data, comparing the two data sets, and suggesting some improvements) then they were awarded at least the pass mark of 40 %. Additional marks were awarded for the level of detail provided, for example, the highest scoring students went on to calculate a discharge coefficient for the system, compared this to published values, identified, and attempted to quantify, sources of experimental error, and also proposed how their suggested improvements could be validated.

2.7. Evaluation of effectiveness

To meet the fourth stage of the IDEE framework, student feedback was collated using online, electronic surveys hosted on the University's VLE. This accessible platform was used due to its suitability of asking multiple choice questions, its familiarity to students, and the ability to easily integrate the surveys alongside the pre- and post-lab work. To maximise response rate, students were required to complete the survey as part of their submissions. The questions asked in each survey are summarised in Table 2 and Table 3, and both surveys included the option to include free-text comments at the end. All questions contained the option "Would rather not say" to give students the choice of whether or not to share their perspectives. Data from all students was downloaded, anonymised, and analysed.

Given the positive orientation of some questions (for example, Pre-lab survey Question 5: "I can see the need for critiquing the output of a large language model"), there is a risk of bias within the descriptive statistics reported in Section 3, where mitigation against this is also discussed. Quantitative data from questions 1–6 in the pre-lab survey (Table 2) and questions 1–5 in the post lab survey (Table 3) were analysed in MS Excel due to the suitability of this software to easily count the number of different responses. Responses for each question were compiled into relevant pie charts to show the response distribution. This is a common approach to reporting nominal data consisting of six or fewer categories and is a widely accepted approach to displaying survey data (Hill, 2024). Qualitative data were imported and analysed in NVivo, where reflexive thematic analysis was used following previously established methods (Braun et al., 2022). As the authors were seeking to understand student perceptions of GAI in an inductive fashion (i.e. without prior expectations), this method was deemed most appropriate for this study. Two authors (Dr. Matthew Keith and Ms. Eleanor Keiller) independently read all qualitative data to familiarise themselves with its content. Codes were independently applied to the data, and then the similarities and differences were discussed between the two authors. A joint coding framework was developed and themes within the codes were identified. All the qualitative data was subsequently re-coded using the agreed coding and thematic framework.

When conducting qualitative data analysis, it is important for researchers to reflexively engage with their professional and personal demographic identities and to understand how this may affect the outcomes that they elicit from the data. For this reason, Walsh's typology of reflexive practices was used throughout the analysis to understand and mitigate potential bias. As a chemical engineering academic and developer of the lab session described herein, the lead author of this work may hold inherent bias and desire to see an improvement in student learning outcomes. It is hoped that the second author (EK), who is an expert in qualitative data analysis within the social sciences, brings a non-subject specific perspective to this process and hence acts impartially.

Table 2
Questions and associated multiple choice answers asked as part of the pre-lab survey.

No.	Question	Possible answers
1	Before university, have you heard of generative artificial intelligence (GAI) such as Chat GPT?	Yes, No, Not sure, Would rather not say
2	Have you used generative AI for any reason before completing the Tank Draining pre-lab work?	Yes, No, Would rather not say
3	Have you used generative AI to help with any previous work?	Yes, No, Would rather not say
4	I believe that generative AI will be useful in helping with future work.	Yes definitely, Likely to be helpful, Not likely to be helpful, No definitely not, Would rather not say
5	I can see the need for critiquing the output of a large language model	Strongly agree, Agree, Neutral, Disagree, Strongly disagree, Would rather not say
6	I understand how to critique the output of a large language model	Strongly agree, Agree, Neutral, Disagree, Strongly disagree, Would rather not say
7	Use this space to include any additional comments on your experience with generative AI to date. You may wish to use this as a point of reference when reflecting on your learning as part of this module.	Free text entry

Table 3
Questions and associated multiple choice answers asked as part of the post-lab survey.

No.	Question	Possible answers
1	As a result of the tank draining lab and associated lecture, my understanding of how to use generative AI in engineering problems has improved	Strongly agree, Agree, Neutral, Disagree, Strongly disagree, Would rather not say
2	As a result of the tank draining lab and associated lecture, my understanding of how to use generative AI ethically has improved	Strongly agree, Agree, Neutral, Disagree, Strongly disagree, Would rather not say
3	As a result of the tank draining lab and associated feedback, my understanding of how to critique the output of large language models (such as ChatGPT) has improved.	Strongly agree, Agree, Neutral, Disagree, Strongly disagree, Would rather not say
4	As a result of the tank draining lab and associated lecture, I now understand some of the limitations of generative AI	Strongly agree, Agree, Neutral, Disagree, Strongly disagree, Would rather not say
5	I am likely to use generative AI to help with future work.	Strongly agree, Agree, Neutral, Disagree, Strongly disagree, Would rather not say
6	Do you have any additional comments you would like to provide based on the tank draining lab and use of generative AI? You may wish to refer back to any thoughts you had when completing the pre-lab survey.	Free text entry

3. Results and discussion

3.1. Experimental and model data

Following the completion of the experiment, example experimental data was made available on the University's VLE. This is shown, along with the output of the model from Eq. (12), in Fig. 3. The discharge coefficient was taken as 0.97 (Hicks and Slaton, 2014).

As illustrated, there was a high degree of repeatability between experimental runs; both tanks showed a percentage difference between runs of less than 2 %. There was, however, some difference in the suitability of this model to represent the two different experimental systems. The larger Tank 2 was very close to the model with a difference of just 0.3–1.4 %. However, the difference between the model and Tank 1 was 10.0–12.2 %. The reason for the difference in the model suitability may be due to the tank geometry; Tank 1, with an inside diameter of 65 mm, has a larger surface area to volume ratio than Tank 2 (inside diameter of

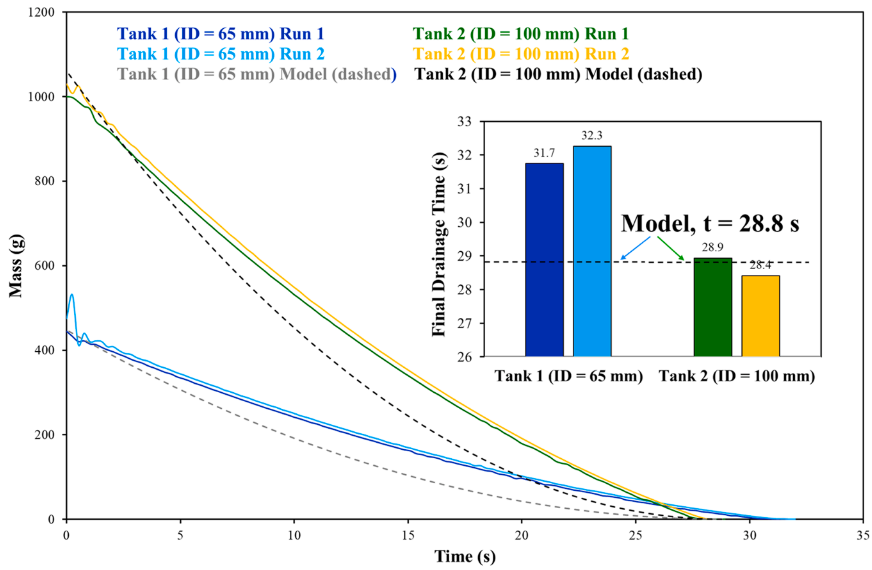

Fig. 3. Experimental and model data for the mass of water in each tank as it drained under gravity.

100 mm). For this reason, there is likely to be some additional effects of friction between the tank wall and the velocity of the fluid level moving downwards. It is also worth noting that the discharge coefficient is dependent on hole shape, relative size, type of edge, and flow profile of the fluid. Although the two tanks and discharge holes were designed to be geometrically similar, there may have been differences in the manufacturing of the plugs. Recalculating a discharge coefficient based on the mean of the experimental data gave a value of 0.921 for Tank 1 and 0.978 for Tank 2. These relatively high values (close to 1) are characteristic of smooth, rounded edged orifices and are similar to those reported elsewhere (Hicks and Slaton, 2014).

Although the experimental and model drainage times are generally quite similar, the drainage profile does show some significant deviation, with the model predicting a faster rate than that measured, particularly between 5 s and 20 s. This may be due to the model not considering the change in height of the fluid, and the subsequent influence that this has on the exit velocity. As the fluid height reduces, there is a smaller pressure head at the outlet, and so the fluid velocity at the outlet will slow down. Secondly, the model does not specifically consider the impact of surface tension at the outlet, although this may be captured in the measured discharge coefficient.

3.2. Assessment of student performance

A similar lab activity was delivered in the 2022–23 academic year without the use of LLMs. In this, students scored a mean mark in the pre- and post-lab work of 57.6 % and 58.0 % respectively. In 2023–24 where ChatGPT v3.5 was incorporated, the mean mark rose slightly to 59.3 % in the pre-lab work, but remained exactly the same in the post-lab sheet as illustrated in Table 4. A two-sample *t*-test conducted in Minitab Statistical Analysis software (v2018.1.0.0) also highlighted that there was no statistically significant difference between the marks achieved by students in 2022–23 and 2023–24 for both the pre-lab and post-lab work. However, it is worth noting that marks were awarded based on slightly different criteria; in 2022–23, the pre-lab was assessed based solely on students' ability to derive a model themselves, whereas in 2023–24, marks were awarded also based on their critique of the LLM output. However, the post-lab exercise remained the same between the two years and so can be directly compared. The similarity in performance between the two groups suggests that the incorporation of ChatGPT v3.5 has not changed students' ability to analyse and process the data nor developed a deeper understanding of fluid dynamics. This is contrast to a recently published meta-analysis of 28 articles which showed that GAI significantly improved academic performance (Sun and Zhou, 2024). With regards specifically to model development and coding, GAI has also helped improve student performance in some work, although there was a concern that there may be an over-reliance on these tools (Dani, 2024). This improvement, however, is not universal and other research has found that students using GAI on average scored 6.7 % lower than those who did not use it for essay-based assignments (Wecks et al., 2024). The educational activity described and evaluated in this paper may, therefore, be improved by providing greater focus on student support, with the possible implementation of small-group based teaching and discussion to facilitate their critique of an LLM output and subsequent data analysis tasks. Although there is not yet a clear improvement in student learning outcomes, a core aim of this activity was to expose students to an LLM, its benefits, and potential pitfalls in an educational setting. Whether students' thought that this was worthwhile is explored in the following sections through the analysis of their perceptions, as obtained in the pre- and post-lab surveys.

Table 4
Mean and standard deviation of student marks achieved in 2022–23 (without use of ChatGPT v3.5) and in 2023–24 (with ChatGPT v3.5).

	Year	Mean	Standard deviation
Pre-lab	2022–23	58.7 %	10.8 %
	2023–24	59.3 %	9.1 %
Post-lab	2022–23	58.0 %	10.3 %
	2023–24	58.0 %	13.2 %

3.3. Pre-lab survey

The pre-lab survey aimed to gain an understanding of students' current knowledge and perception of GAI. The responses to Questions 1–4 are provided in Fig. 4, with nearly all people choosing to respond. This shows that the vast majority of students (92 %) have heard of GAI technologies, whilst nearly half (44 %) had used it for something prior to completing the pre-lab work. This is concomitant with previous reports published at a similar time to this survey which showed that many students, studying a range of subjects, are increasing their familiarity with AI tools (Faruk et al., 2023; JISC, 2023; Johnston et al., 2024). A significant proportion also admitted to using GAI for work-related purposes in the past; although this 22 % is still in the minority, 81 % believed that GAI will be "useful in helping with future work". With the

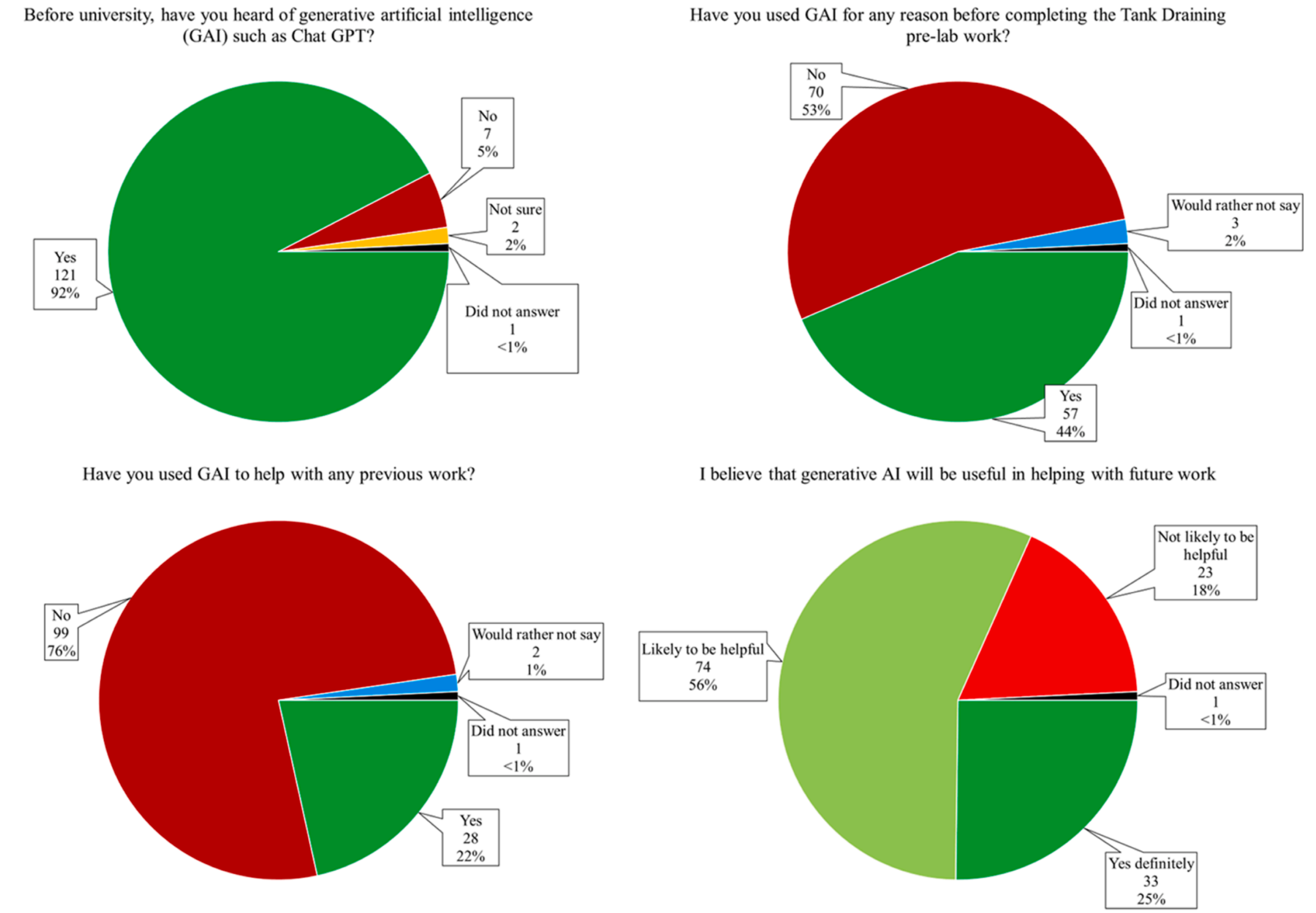

Fig. 4. Pre-lab survey responses to questions 1–4 assessing current knowledge, understanding, and use of GAI (n = 130).

further integration of AI tools into existing, widely used programmes (e.g. the roll-out of Microsoft Copilot across Office applications (Carter, 2023)) this figure is likely to grow over the coming years. Indeed, a significant growth in acceptance amongst University students from 2023 to 2024 has already been highlighted (JISC, 2024). This growth in usage has been highlighted as both a potential concern and opportunity (Alier et al., 2024), with various frameworks published to ensure that AI can be integrated effectively into higher education (Holmes and Miao, 2023; Su and Yang, 2023).

Before the lab session, the majority of students (92 %) either strongly agreed or agreed that there was a need to critique GAI output, as shown in Fig. 5. Response bias, due to how the question was phrased, should also be considered here. This means that there is a higher proportion of people strongly agreeing, or agreeing, with the need to critique the output, even though they might not actually do so. Although this may have been a leading question asked after the lecture where limitations and pitfalls were discussed, previous work has highlighted that students across different courses have concerns regarding the accuracy of LLM outputs (Chan and Hu, 2023). If it is so widely agreed that the output may not be trustworthy, this may allay some fears of faculty members

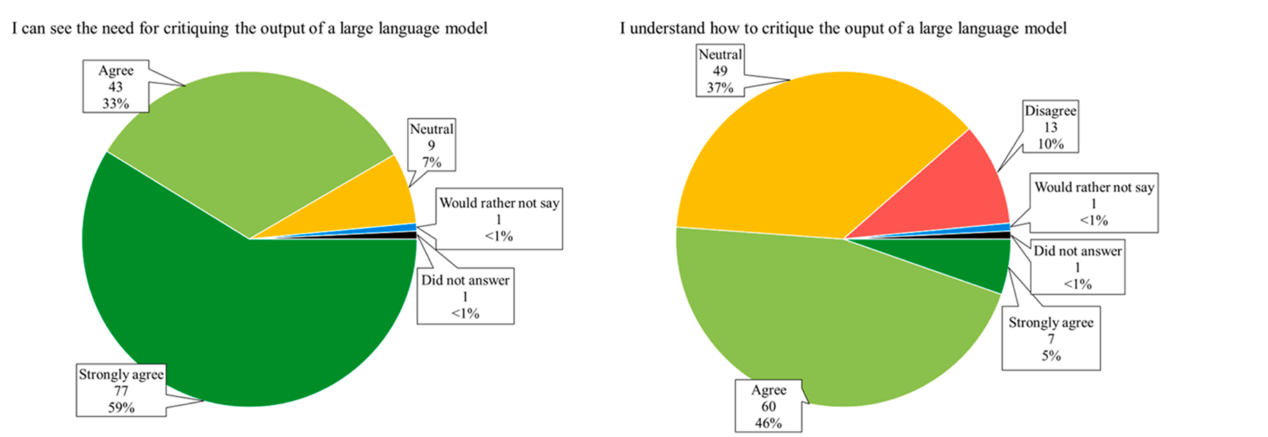

Fig. 5. Pre-lab survey responses to questions 5 and 6 assessing students' perception of the need for critiquing and knowing how to critique the output of GAI (n = 130).

that LLMs will be used for plagiarism and cheating on coursework assignments (Alier et al., 2024; Cotton et al., 2024). Indeed, additional work has found that students themselves are also mindful that over-use of LLMs may violate academic integrity (Johnston et al., 2024).

Interestingly, before the lab session, approximately half of students strongly agreed or agreed that they understood how to critique the output of an LLM. Response bias should also be considered here as there may be a lack of willingness from people to admit that they do not know how to critique the output effectively. This may explain the wide spread of marks shown in Fig. 6 where both the lowest and highest marks of 20 % and 75 % were given to people who agreed that they knew how to critique the output. Note that this section was marked out of 20, with up to 50 marks available for the pre-lab work. These marks are indicative of whether an individual was able to engineer input prompts (LO1), critically evaluate the output (LO2), suggest improvements (LO3), and coherently articulate their arguments, with most marks awarded for the quality of their critical evaluation. As illustrated, there is often a disparity between a student's initial perception of whether they can critique the output of an LLM and their ability to demonstrate that they can do so. There is a similar spread in marks of people who answered "Neutral" to those who "Agreed" with this statement, suggesting that there may also be a lack of confidence amongst students about using GAI for work-related purposes, particularly with regards to preserving academic integrity (Johnston et al., 2024). Both findings highlight that there is a need for effective education on the use of GAI within a wide range of University degree programmes, which is becoming more prevalent around the world (Hashmi and Bal, 2024). A recommendation for future educators regarding the implementation of GAI into their courses is therefore, to provide specific education on academic integrity, including the use of recognised referencing styles.

3.4. Post-lab survey

Following the completion of the lab session and the return of feedback on the pre-lab worksheet, a follow-up survey was conducted. The number of responses (121) was fewer than the responses received from the pre-lab survey as some students no longer studied the course. This survey aimed to understand how students' perceptions of GAI had

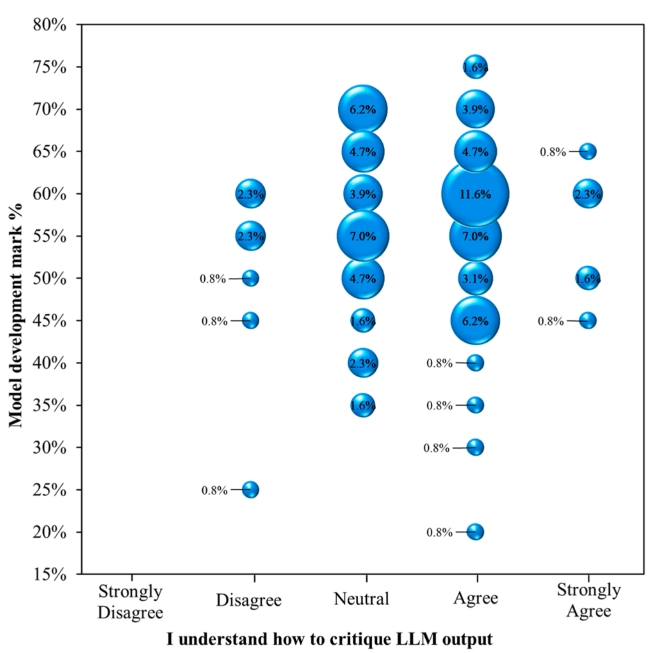

Fig. 6. Marks awarded for developing a model with GAI (including critique of output) compared to whether a given student thought they understood how to critique an LLM.

changed as a result of the lecture, lab session and feedback, and whether they thought that they had met the intended learning outcomes. The aggregated responses to this survey are provided in Fig. 7 and Fig. 8. Response bias must again be considered here; students may be likely to answer positively to these questions as there may be a desire to show that they have met the learning outcomes of this work. Similarly, in Fig. 8, there may be a hesitancy to admit their true feeling regarding their likelihood of using LLMs for future work, in case they are later accused of plagiarism.

With the limitations described above in mind, the majority of students reported that, as a result of the teaching and learning activities described in Section 2, their ability to use GAI ethically in engineering problems had improved (LO1). Similarly, most students felt that they now understood some of the limitations of LLMs, such as ChatGPT v3.5. As one of the aims of this lab session was to illustrate some of the pitfalls of GAI (LO2), the educational activities described herein may be regarded as a successful way of doing this. However, as Fig. 6 shows, there is a disconnect between some students' perceived skills and their ability to demonstrate them. This is mirrored in previous work which involved a cross-discipline study of over 1000 students and found that there is a risk that students' confidence in using GAI may be overstated (Kelly et al., 2023). Addressing this risk may be a key consideration for other educators. Hence, a follow up activity allowing students to showcase their learning represents an opportunity for this lab session to be improved. This future activity could focus on students demonstrating their understanding of AI behaviour and its limitations (e.g. a lack of logical reasoning and inability to understand the text which it generates). In turn, this will foster the responsible use of AI; a crucial skill for future engineering graduates (Marcus, 2018). It will also mitigate against the potential response bias of the survey questions by allowing students to demonstrate, and further develop, their AI literacy.

It is also worth noting that there is a small proportion of people who reported not gaining an improvement in their AI skills. Unfortunately, the majority of these did not provide free-text comments and so it is difficult to understand why this is the case, although one person did state that "clearer guidance on directing chat-gpt prompts…would have been helpful". Based on this, some additional examples can be provided in future lectures and thus aid students in meeting LO1. This may be particularly helpful in designing future teaching activities, especially since it has been previously recommended that the skill of "prompt engineering" is developed as part of an AI-enabled future (Hashmi and Bal, 2024).

Although most students felt that their understanding of GAI had improved, the response to whether they were likely to use this tool in future work was much more mixed. As shown in Fig. 8, a minority (27 %) either strongly agreed or agreed that they were likely to use GAI in future work. However, comparing responses in the pre- and post-lab surveys, as shown in Fig. 9, reveals an interesting result. Here, 80 % of respondents believed in the pre-lab survey that it was at least "likely to be helpful" in future work, yet 70 % of people were either neutral or unlikely to use it in future after completing the lab session. As noted previously, this may be due to a lack of willingness to admit to its use, or it may be that this activity effectively demonstrated its limitations and hence discouraged using GAI. Although this latter point may be beneficial with regards to preserving academic integrity, it may also be an unintended consequence of this lab session which, in part, aimed to develop skills in the responsible, ethical, and effective use of AI. An alternative explanation may be that students' perception is that, the extra work involved in prompt engineering, fact-checking, and critiquing outweighs the potential benefits of being able to produce large volumes of text. It is worth emphasising that, as a tool, the use of AI is likely to grow, and so educators have a responsibility to ensure that future engineers are equipped with these skills. A major consideration for higher education providers is that with such abundant access to information, the value and purpose of a degree is likely to move from a graduate merely possessing advanced technical knowledge to having the

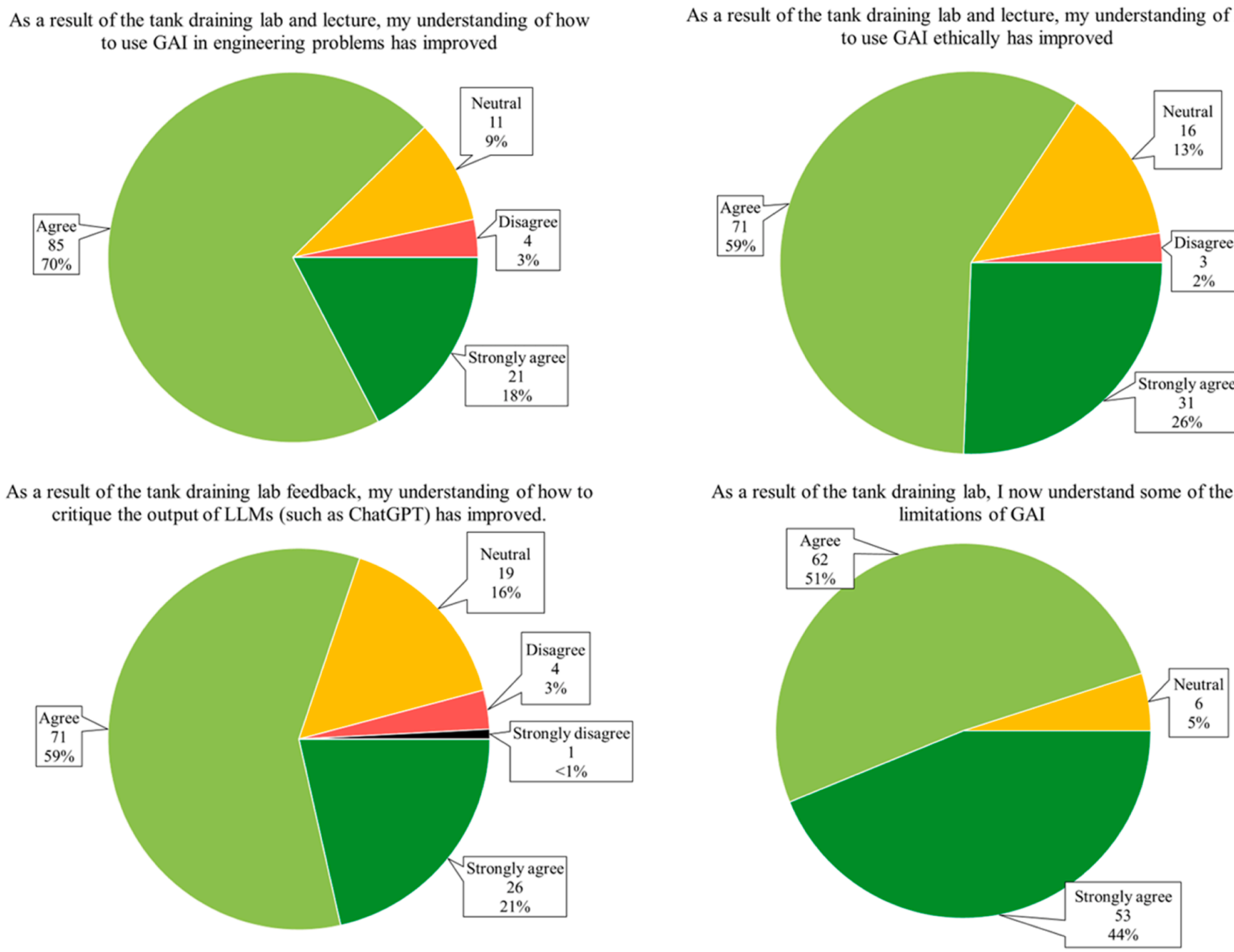

Fig. 7. Post-lab survey responses to questions 1–4 assessing students' perceptions of how their understanding and use of GAI has changed as a result of this lab session (n = 121).

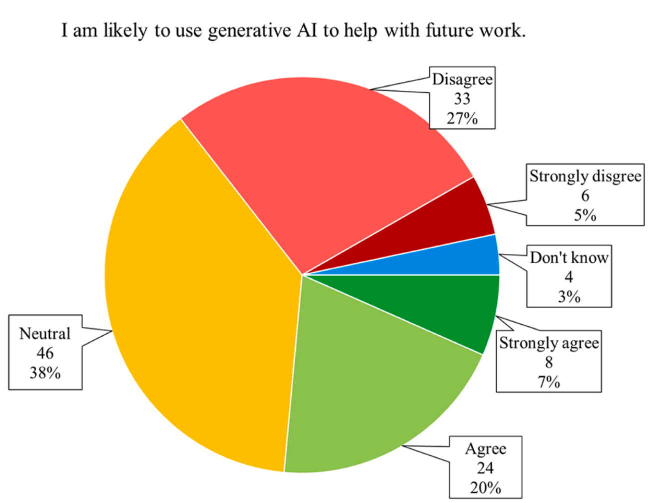

Fig. 8. Post-lab survey response for question 5 assessing how likely students are to use GAI in future work (n = 121).

transferrable skills of critical thinking, emotional intelligence, and ethical judgement. The development of such skills can hence complement the advent of GAI and mean that future graduates are unlikely to be replaced by it (Hashmi and Bal, 2024).

3.5. Thematic analysis

Reflexive thematic analysis (a systematic approach to analysing qualitative data to identify common themes, described in detail elsewhere (Braun et al., 2022)) was applied to the free text comments collated following the completion of the pre- and post-lab surveys. As a result of this, 27 independent codes were developed inductively (i.e. driven by the data) which informed eight distinct themes. These codes and corresponding themes are provided in Table 5.

The first theme elicited from the analysis was that many students had limited prior experience with GAI such as ChatGPT. Although some had used it recreationally, such as for social or creative projects, many had not used it at all. This reinforces the data shown in Fig. 4 and, as a relatively new technology, perhaps this theme is not that surprising. It is also concomitant with other literature which often suggests that there is a mixture of different experiences regarding current usage of GAI (Johnston et al., 2024). Interestingly, Theme 2 suggests that there is a limited intention of students to use GAI in future work. Although this does not completely preclude LLMs being used, many comments stated that it should be a "last resort", that "we shouldn't rely heavily on it", and that "there is a chance of generative AI giving the wrong information and it being unethical". This is in some contrast to Fig. 4 where 81 % said that GAI was either definitely or likely to be helpful in future work, although it does confirm the finding in Fig. 8 where, in the post-lab survey, the likelihood of students using it in future work was highly variable across the cohort.

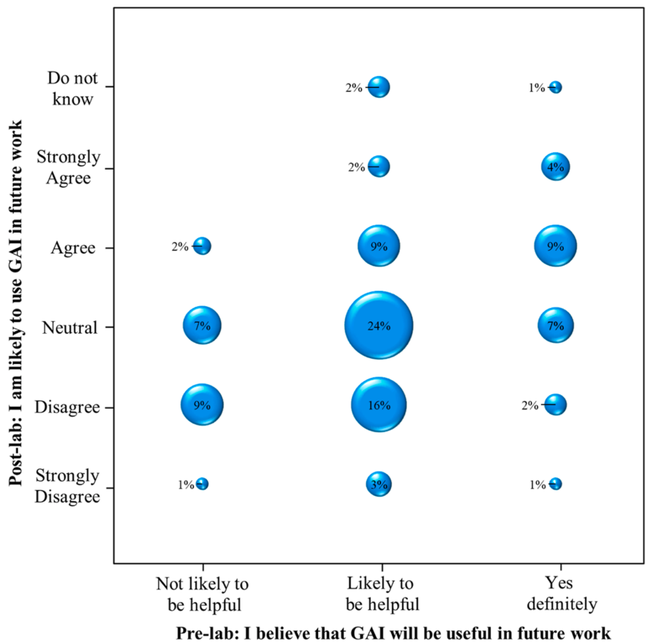

Fig. 9. Post-lab likelihood of using GAI versus pre-lab perception of usefulness of GAI.

Despite this apparent intention to limit the use of GAI, some students did report some benefits of using it, as identified in Theme 3. Some people recognised its potential in supporting their education particularly noting that it conveniently consolidates information for them; for example, one comment stated that "Chat GPT is useful in gathering lots of information". Similarly, some students also noted that GAI can support their understanding of certain topics, theories, or programmes, and that it can be a useful starting point. Indeed, many were able to effectively use the model to develop similar plots to that provided in Fig. 3, thus demonstrating LO3 had at least been partially met. Similar benefits have also been highlighted in previous work exploring the disruptive effect of GAI (Alier et al., 2024) and student perceptions of this technology (Chan and Hu, 2023; JISC, 2024). Indeed, students from a range of disciplines including Arts, Business, Dentistry, Education, Law, and Social Sciences have highlighted some additional and complementing benefits to those reported in this research. These include the provision, by ChatGPT, of personalised and immediate learning support, writing aid (especially for non-English speakers), and the automation of repetitive tasks (Chan and Hu, 2023).

One respondent also noted that they "had to make sure they [the outputs] were accurate". This was a commonly occurring theme which was elicited from the data, and so Theme 4 was defined as perceived disadvantages of GAI. The perception that GAI "caused confusion", resulted in "more work", and "was difficult" to use was also prevalent, which may hinder the achievement of LO1. These findings highlight two important aspects with regards to this teaching exercise. The issue of confusion should be addressed in future iterations of this teaching exercise, perhaps through the release of additional resources following the completion of the pre-lab work. Secondly, the difficulty associated with using GAI might be overcome through a computer-based tutorial session in addition to just demonstrating the use of Chat GPT v3.5 as part of the lecture. This approach of developing tutorials in support of problem-based learning exercises has been shown to raise student competency and improve learning outcomes (Binani and Chowdary, 2018; Ekabote and Kodancha, 2015). It has also been previously highlighted that students should be explicitly taught the use of GAI (Kelly et al., 2023), and so there is a potential opportunity to develop LLM-focussed computer-based classes. This is supported by additional work exploring the impact of GAI on University policy, with recommendations being that

Table 5
Codes, associated themes and example free text comments provided in pre- and post-lab surveys.

#	Code	Example comment	Theme
1	Not used before	Response 33: "I was relatively inexperienced with generative AI before this, and therefore did not see how it could be used effectively"	1. Students had limited prior experience with Generative AI
2	Does not meet expectations	Response 28: "After the pre-lab sheet I have realised the it isn't as powerful as it is made out to be."	
3	Should not become reliant on GAI	Response 20: "We can use it from time to time for convenience's sake, but we shouldn't heavily exploit and rely on it."	2. Limited intention for the use of GAI
4	Should not be used for science	Response 13: "If not necessary, DO NOT USE FOR SCIENTIFIC WORK"	
5	Last resort use	Response 3: "ChatGPT should be the last resort."	
6	Discouraged to use ChatGPT	Response 24: "Thas discouraged me from using it [ChatGPT] as I could not manage to get ChatGPT to output the correct equation. I think it got close but what is the point in using it if it is not correct and does not know if it is wrong?"	
7	Aid understanding	Response 3: "ChatGPT can help when you do not understand what a question is asking or the solution."	3. Perceived benefits and uses of GAI
8	Convenient tool	Response 41: "I think that Chat GPT is useful in gathering lots of information into one website."	
9	Useful / helpful tool	Response 91: "I found the outputs to be very useful, but had to make sure they were accurate. It gave me a good place to start my model description."	
10	Generates ideas	Response 10: "Generative AI has been helpful in the innovation of ideas"	
11	Takes time to correct mistakes	Response 13: "More time will be spent correcting and critiquing it than the time needed to do the work manually."	4. Perceived disadvantages of GAI
12	Can cause confusion	Response 8: "[Using GAI] leads to confusion on what the "true" answer is and what to base my analysis off"	
13	Difficult to use	Response 23: "The AI was difficult to get to do what I wanted it to do."	
14	Tasks are more difficult with GAI	Response 96: "I have never used AI before the Tank Draining Pre-Lab work, and since it does not give me the output that I wanted, it was more work for me than when I have to search it up manually"	
15	Not always right	Response 9: "The output from ChatGPT it is often wrong and it isn't always easy to check."	5. Perceived quality of the outputs of GAI

(continued on next page)

Table 5 (*continued*)

#	Code	Example comment	Theme
16	Not reliable	Response 16: "I believe the AI is not that reliable to do the task on its own."	
17	Untrustworthy	Response 25: "This was my first time using generative AI. I did not like it, It felt untrustworthy with my data and inaccurate with calculations."	
18	Inconsistent	Response 15: "So far it has been inconsistent and feels like a lottery."	
19	Ignores mathematical logic	Response 14: "In addition, I think generative AI cannot be used for calculation, because it may ignore some mathematical logic, in turn the result is inaccurate."	6. Awareness of the pitfalls of GAI
20	Limited function	Response 10: "Generative AI has been helpful in the innovation of ideas but becomes limited when needed for more complex activities"	
21	Need high quality input prompts	Response 25: "If the prompt has enough data and detail, a reasonable output can be produced."	
22	Critical assessment of outputs	Response 9: "When critically analysing the output from ChatGPT it is often wrong and it isn't always easy to check."	7. Awareness of the need for critical evaluation of GAI
23	"Careful" use of GAI	Response 40: "I enjoyed using Chat GPT as part of the work and I think it could be a useful tool in the future if properly managed. I did find it useful despite its pitfalls but you have to be very careful when using it."	
24	Difficulty in critiquing	Response 18: "I believe with more practise, I hope I'll get better at spotting its technical critiques and get a better understanding of the overall AI based lab reports like this."	
25	Used alongside own / existing knowledge	Response 32: "If I didn't have prior knowledge in fluid mechanics there would have been big problems with comprehension."	
26	Sense of danger / fear	Response 32: "ChatGPT is very useful but at the same time very dangerous." Response 12: "I've never used it before, but I find the idea of it taking over jobs intimidating."	8. Feeling and emotion associated with GAI
27	Strangeness	Response 18: "I found AI particularly clever, but also quite odd in the way it talks. To be honest, I found it hard to see if had any errors in the answers, but just the way it provides answers was weird"	

higher education institutes provide highly prescriptive guidelines (Malik et al., 2023).

Closely related to these experienced disadvantages of using GAI, is the perceived quality of the output of LLMs which has been captured in Theme 5. In particular, students noted that the output was "often wrong", "not that reliable", and "inconsistent". Although demonstrating this was not a specific aim of the teaching activity, highlighting the possible inconsistency of LLMs, along with reinforcing that they can be susceptible to providing misinformation (Chen and Shu, 2024), is thought to be an additional benefit. One of the main aims of this activity was to provide a learning opportunity so that students could experience some of the limitations and pitfalls of GAI. Theme 6, which has been informed by the codes "Ignores mathematical logic", "Limited function", and "Needs high quality input prompts", directly addresses this aim. This is related to, but distinct from, Theme 5; whereas Theme 5 considers the end result (or output) of ChatGPT v3.5, Theme 6 is more concerned with the flaws in how it works. Many students noted that it did not use proper mathematical logic, that it could not accurately complete an integration, and failed to set up the initial problem in a logical manner. Some students took the identification of these flaws one step further and identified that ChatGPT v3.5 has "limited function". Finally, students often realised that there is a strong need for a clear, high-quality input in order to obtain a meaningful output, thus addressing LO1. This ability of "prompt engineering" has been previously highlighted as a necessary skill in a future AI-enabled workplace (Tsai et al., 2023) while other research has shown that LLM performance is improved by providing high-quality inputs, such as those generated using effective prompt patterns (White et al., 2023). Incorporating research such as this into an AI-based tutorial (as suggested above) may also enable students to develop their prompt engineering skills.

Perhaps due to the limitations and pitfalls of GAI, Theme 7 showed that students were often aware of the need to critically evaluate the output. The importance of being "careful" and using their "own knowledge" to confirm the output was often noted. Demonstrating this, and giving students the opportunity to learn how, was a key aim of the lab session. This is reinforced by the quantitative survey data provided in Fig. 7 which assessed the perceived change in understanding how to critique LLM output. Further pedagogical research could be carried out in this area through setting an additional AI-orientated assignment after completing this teaching activity and subsequently investigating the change in students' critical analysis. This approach of step-wise knowledge building and increasing student exposure to AI-driven tools and techniques has also been described in other research (Lee et al., 2023).

Finally, Theme 8 centred on the feelings and emotion associated with the use of GAI. Generally, these were not positive and the emotions elicited were a sense of danger, fear, or intimidation, or that GAI was simply "weird". This was just a small aspect of the data, but an important one nonetheless. Developing an understanding of how students feel about the use of GAI, particularly for summative assignments, may inform future education and policy in this area (Malik et al., 2023). One concern was that the "idea of it taking over jobs [is] intimidating". To address this, the focus within higher education therefore needs to be on AI literacy, such that graduate students are able to use AI effectively, and apply independent critical analysis, ethical judgement, and emotional intelligence (Bearman and Ajjawi, 2023). Communicating the development of these skills, and techniques such as reflective practice, will also be essential so that students develop the self-awareness needed in order demonstrate their abilities in future graduate employment.

4. Conclusion

This work has described and evaluated a method for integrating GAI, specifically LLMs into chemical engineering curricula. The primary teaching activity was developed using the principles of problem-based learning. This was also supported by an initial lecture which described the principles of GAI, it's ethical use, and how it can be applied to engineering problems. Following this, students were tasked with developing a model to predict the drainage profile of a tank using the widely accessible platform ChatGPT v3.5. Hence, students were provided with the opportunity to explore the benefits, limitations, and pitfalls of this nascent technology. To explore how well their model reflected reality, students then completed an experimental lab session and compared the data obtained to their model in the post-lab work. As described herein, the model (which used literature values for discharge coefficient) provided a reasonably good prediction of the experimental data, especially for a tank with a larger geometry.

As GAI, and LLMs in particular, are only just being incorporated into higher education curricular, pre- and post-lab surveys were conducted in order to gain insight into students' experiences and perceptions of using this new technology. By carefully analysing the output, comparing this with their own model derived from first principles, and consulting relevant lecture material and published literature, it was hoped that students would meet three distinct LOs. Most people acknowledged that there was a clear need to critically analyse the output, and indeed a significant proportion thought that they already knew how. However, correlating these responses with students' marks for that section of the coursework showed that, often, students were not able to demonstrate this critical analysis ability at a 2(i) or First class standard. Following the lab session and feedback on the pre-lab work, students generally thought that this teaching activity had improved their understanding of how to use GAI and how to use it ethically (LO1), along with skills in critiquing the output (LO2) and identifying improvements (LO3). However, to confirm whether these outcomes had actually been met, a follow-up AI-focussed assignment would need to be completed.

Reflexive thematic analysis was applied to the qualitative data obtained from the free text comments in the pre- and post-lab surveys. This revealed eight distinct themes which generally confirmed the quantitative survey data, namely that students generally had limited prior experience of using GAI, that there were limited intentions with regards to its' future use, and that there was a need to critique the output. However, it also offered a richness beyond the quantitative data and revealed that students were aware of specific pitfalls such as this LLM's ignorance of mathematical logic and need for high-quality input prompts.

As GAI, including LLMs, continue to gain prevalence in both education and workplace settings, it is essential that future graduates are equipped with the knowledge and skills to use it ethically, responsibly, and safely. It is hoped that the lab session described herein provides a valuable case study, exampling how chemical engineering curricula could be augmented to expose students to LLMs. While this is the responsibility of educators, it is the responsibility of students to develop the critical analysis skills to ensure that they are fully literate in future AI-enabled environments. Engagement with practical, problem-based teaching activities such as this provide the opportunity for students to develop these skills.

Author Contributions

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ece.2025.01.002.

Data availability

Additional survey data is available in the Supplementary Materials.

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